

# Linear Algebra Mid-Term Exam

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**Grading Scheme:** If  $w$  is the wrong answer count, and  $c$  is the correct answer count, then your grade on a scale from 0 to 20 will be computed as follows

$$\frac{20 - w + 2 \times c}{60} \times 20 .$$

- 4 points per correct answer
- -1 point per wrong answer
- 0 points per NO RESPONSE

Every question has exactly one correct answer.

Use the grid provided here to declare your answers.

Write using block capital letters, A, B, C *etc.*

If you choose to NOT answer a question, write NO RESPONSE in the blank.

Illegible, ambiguous, or multiple responses per question will be considered incorrect.

- |           |           |
|-----------|-----------|
| 1. _____  | 12. _____ |
| 2. _____  | 13. _____ |
| 3. _____  | 14. _____ |
| 4. _____  | 15. _____ |
| 5. _____  | 16. _____ |
| 6. _____  | 17. _____ |
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| 9. _____  | 20. _____ |
| 10. _____ | 21. _____ |
| 11. _____ | 22. _____ |

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# 1 Object Orientation

1. What does the magic method `__add__` do in a Python class?
  - A. Overloads the `+` operator
  - B. Controls how an object is printed
  - C. Nothing because the method is an `@staticmethod`
  - D. Causes addition to be commutative.
2. Suppose classes `A` and `B` are defined as follows.

```
1  class A:
2      def f(self, s):
3          return "[" + s + "]"
4
5      def g(self, s):
6          return "@"
7
8  class B(A):
9      def g(self, s):
10         return self.f(s)
```

Now what does the expression `[B().g("x"), A().g("x")]` evaluate to?

- A. `["", ""]`
  - B. `["[x]", "@"]`
  - C. `["[0]", "@"]`
  - D. This expression raises an error because `f` is not defined in `class B`.
3. What is a static method?
    - A. A method which does not take an object of the class as first argument.
    - B. A method which always returns `None`
    - C. A method which never returns `None`
    - D. A recursive method

# 2 Vectors

4. The set of vectors,  $v \in \mathbb{R}^3$  together with vector addition is a monoid. What is the identity element of the monoid?

- A.  $1 \in \mathbb{R}$       B.  $0 \in \mathbb{R}$       C.  $\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \in \mathbb{R}^3$       D.  $\begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \in \mathbb{R}^3$       E.  $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \in \mathbb{R}^{3 \times 3}$
- F.  $\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \in \mathbb{R}^{3 \times 3}$

5. Evaluate the following inner product:  $\begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix} \cdot \begin{bmatrix} -1 \\ 2 \\ 2 \end{bmatrix}$

- A.  $\begin{bmatrix} -1 \\ -2 \\ 4 \end{bmatrix}$       B.  $\begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$       C.  $\begin{bmatrix} -1 & 2 & 2 \\ 1 & -2 & -2 \\ -2 & 4 & 4 \end{bmatrix}$       D. 1      E. 0

6. Let  $u, v, w \in \mathbb{R}^n$ , and let  $s, t \in \mathbb{R}$ . Which of the following is not necessarily true?

- A.  $u + v = v + u$   
 B.  $s(u + v) = su + sv$   
 C.  $u(v + w) = uv + uw$   
 D.  $(s + t)(u + v) = su + sv + tu + tv$
7. Let  $v = \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}$ . Which scalar  $s \in \mathbb{R}$  satisfies  $sv = \begin{bmatrix} -2 \\ 2 \\ -2 \end{bmatrix}$ ?  
 A.  $s = 0$    B.  $s = 1$    C.  $s = -1$    D.  $s = -2$    E.  $s = 2$
8. Given two vectors  $u, v \in \mathbb{R}^5$ , we know that  $u + v = v + u$  because  
 A. vector addition is commutative  
 B. vector addition is associative  
 C. vector addition is invertible  
 D. vectors are scalable
9. What is the cosine of the angle between the vectors  $u = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$  and  $v = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$ ?  
 A. 0   B.  $\frac{1}{\sqrt{3}}$    C.  $\frac{2}{\sqrt{3}}$    D.  $\frac{2}{\sqrt{6}}$
10. What is  $\|u\|$  if  $u = \begin{bmatrix} 1 \\ 2 \\ -1 \\ \sqrt{3} \end{bmatrix}$ ?  
 A. 0   B.  $\sqrt{3}$    C. 3   D. 9
11. The following is a buggy implementation of the `angle` method in the `Vector` class. One bug is that if either `u` or `v` is the zero vector, then a `ZeroDivisionError` will be thrown. What is the second bug?
- ```

1 class Vector:
2     ... # lines omitted
3
4     def angle(u, v):
5         assert isinstance(v, Vector)
6         assert u.dim == v.dim
7         import math
8
9         tmp = u.inner_product(v) / (u.norm() * v.norm())
10        return math.acos(tmp)

```
- A. The code will fail because there is no `self` parameter defined.  
 B. There is no termination condition for the recursion  
 C. The method is missing the `@staticmethod` decoration  
 D. A `ValueError` exception will be raised if `tmp` is greater than 1 or less than -1.

### 3 Matrices

12. Suppose  $A$  and  $B$  are both  $3 \times 3$  matrices,  $A, B \in \mathbb{R}^{3 \times 3}$ , such that  $AB = 0$ , where  $0$  is the  $3 \times 3$  zero matrix. Which of the following is true?
- A.  $A = 0$  or  $B = 0$
  - B.  $A = 0$  and  $B = 0$
  - C.  $A = B^{-1}$
  - D. Neither  $A$  nor  $B$  is invertible.
13. Assuming  $A$  is a square, invertible matrix,  $A^T$  is the transpose of  $A$ ,  $I$  is the identity matrix with same dimension as  $A$ , and  $n, m \in \mathbb{N}$ . Which pair of matrices IS NOT multiplicatively commutative.
- A.  $A, A^{-1}$     B.  $A^n, A^m$     C.  $A + I, A - I$     D.  $A + A^{-1}, A - A^{-1}$     E.  $A, A^T$
14.  $A$  is an  $n \times n$  matrix with determinant equal to zero. Which is true?
- A.  $n$  is prime.
  - B.  $A$  is not invertible
  - C.  $A^{-1}$  has a determinant equal to zero.
  - D.  $A$  is equal to its transpose.
15. Evaluate the matrix product:  $\begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & -1 \end{bmatrix} \times \begin{bmatrix} 1 & 2 \\ -1 & 2 \\ -1 & 1 \end{bmatrix}$
- A.  $\begin{bmatrix} -4 & 9 \\ 1 & 0 \end{bmatrix}$     B.  $\begin{bmatrix} 1 & 4 & -1 \\ -1 & 0 & 7 \\ -1 & -1 & -5 \end{bmatrix}$     C.  $\begin{bmatrix} 9 & -4 \\ 0 & 1 \end{bmatrix}$     D. The product does not exist.
16. Suppose  $A \in \mathbb{R}^{2 \times 3}$ ,  $B \in \mathbb{R}^{3 \times 4}$ ,  $C \in \mathbb{R}^{4 \times 2}$ . What is the dimension of  $ABC$ ?
- A.  $2 \times 2$     B.  $3 \times 3$     C.  $4 \times 4$     D. We cannot know without knowing the values of  $A$ ,  $B$ , and  $C$ .
17. Given  $A, B, C \in \mathbb{R}^{3 \times 3}$ , which of the following holds?
- A.  $AB + BC = CB + BA$
  - B.  $AB - BA = 0$
  - C.  $(A + B)C = A(B + C)$
  - D.  $(A + B)(B + C) = AB + AC + B^2 + BC$
18. What is the determinant of  $\begin{bmatrix} 0 & 3 & 0 & 0 \\ 1 & 1 & 2 & 3 \\ 4 & 5 & 5 & 6 \\ 7 & 7 & 8 & 9 \end{bmatrix}$ ?
- A. 0    B. 1    C. -3    D. 6830

## 4 Gaussian Elimination

19. Given  $A = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 2 & 2 & 1 \end{bmatrix}$ . Perform the following row operations, assuming the rows and columns use 1-based indexing; *i.e.*, the top row is  $R_1$ , not  $R_0$ .

$$2R_2 - R_3 \rightarrow R_3$$

$$-R_1 + R_2 \rightarrow R_2$$

At this point which row operation will result in the identity matrix?

A.  $3R_2 - 2R_3 \rightarrow R_3$

B.  $-R_3 \rightarrow R_3$

C.  $-R_2 \rightarrow R_2$

D.  $R_2 \leftrightarrow R_1$

20. What is the value of `x` after this code executes?

```
1 m = Matrix([[1, 1, 1, 1], [1, 1, 1, 0], [1, 1, 0, 0]])
2 m = m.swap_rows(0,1)
3 m = m.row_combination(1, 1, -1, 2)
4 m = m.row_combination(-1, 2, 1, 1)
5 m = m.transpose()
6 x = m[0]
```

- A. 0   B. 1   C. (1, 1, 0)   D. (1, 0, 0)

21. You would like to perform Gaussian elimination or Gauss-Jordan elimination, but there is a zero upper left position in the matrix. You need to swap that row with a pivot row. To find the best pivot row, we should

A. search for a 1 in the left-most column.

B. search for the largest absolute value in the left-most column.

C. scale the first row by a constant equal to the maximum element in the column.

D. return `None`, because the matrix is not invertible.

22. Suppose  $x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$ ,  $A = \begin{bmatrix} 1 & 2 & 3 \\ -1 & 2 & -1 \\ 2 & 1 & 5 \end{bmatrix}$ , and  $b = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$ . Given that  $\det(A) = 2$ . And given

$$\det\left(\begin{bmatrix} 1 & b_1 & 3 \\ -1 & b_2 & -1 \\ 2 & b_3 & 5 \end{bmatrix}\right) = -7. \text{ What is the value of } x_2?$$

A.  $\frac{-7}{2}$

B.  $\frac{-2}{7}$

C.  $-7 + 2 = -5$

D.  $-7 \times 2 = -14$